

Team default - GFP Design Rationale

2026 Protein Design in SynBio Challenges

Team Strategy Overview

Design Philosophy: Surface charge engineering + Progressive scaffold optimization + Cross-species chimera

- A. Surface charge engineering -- TGP-inspired Glu enrichment to reduce aggregation (G2, G7)
- B. Progressive stabilization -- Stepwise addition of FoldX-verified mutations (G3, G4)
- C. Brightness combo optimization -- Multiple brightness-enhancing mutations on brightest scaffold (G9)
- D. Cross-species chimera -- Hybrid combining best features from multiple GFP species (G11)

Design 1 - sfGFP_Plus_v2_TGP (default_1)

Scaffold: sfGFP (PDB: 2B3P, 238 aa) | WT DeltaG: +3.40

Strategy: TGP-inspired surface charge engineering + core stabilization

Mutations: Q69L, K79E (TGP surface Glu), N149K, N164Y, E172K, Q204E (TGP surface Glu)

Logic: TGP (Close 2015) showed surface Glu engineering dramatically increases thermostability. K79E and Q204E add negative surface charges. Combined with core stability mutations for dual stabilization.

Design 2 - amacGFP_Plus_v1 (default_2)

Scaffold: amacGFP (PDB: 7LG4, 238 aa) | WT DeltaG: +3.55 | Brightness: 3.97

Strategy: Minimal modification -- only 3 high-impact FoldX mutations

Mutations: Y39N (DDG=-4.36), N149K (DDG=-4.06), E172K (DDG=-3.06)

Logic: Most conservative design. 3 mutations each with DDG < -3.0. Y39N alone turns amacGFP stable (DeltaG=-0.81). Minimal perturbation preserves bright chromophore microenvironment.

Design 3 - amacGFP_Plus_v2 (default_3)

Scaffold: amacGFP (PDB: 7LG4, 238 aa) | WT DeltaG: +3.55

Strategy: Moderate optimization -- 7 mutations: FoldX + usGFP core

Mutations: Y39N, Q69L (usGFP), S147P, N149K, N164Y (usGFP), E172K, D190N

Logic: Intermediate design bridging minimal (default_2, 3 mut) and maximum (32-603_2, 8 mut). All 7 independently verified as stabilizing. usGFP core packing adds hydrophobic core efficiency.

Design 4 - ppluGFP_Opt_v2_surface (default_4)

Scaffold: ppluGFP (PDB: 2G6X, 222 aa) | WT DeltaG: -49.88 | Brightness: 4.23

Strategy: Surface Glu engineering on naturally ultra-stable scaffold

Mutations: S73E, S147E, S174E (surface Glu), L206E (brightness+surface), Y88F, S148A, R211F

Logic: ppluGFP is already most stable (DeltaG=-49.88). Following TGP paradigm, we add 3 surface Glu residues for anti-aggregation. Explores whether surface engineering can further improve already exceptional stability.

Design 5 - cgreGFP_Opt_v2 (default_5)

Scaffold: cgreGFP (PDB: 2HPW, 235 aa) | WT DeltaG: -26.89 | Brightness: 4.50

Strategy: Brightness combo -- multiple verified bright mutations

Mutations: K8Q (N-term stability), K167M (4.57, brightest mutant), D170E (4.56), L197P

Logic: Pure brightness optimization on already-brightest-and-stable scaffold. K167M is single highest-brightness mutation in dataset (4.57). Combo targets chromophore microenvironment from multiple angles.

Design 6 - Chimera_sf_amac (default_6)

Scaffold: sfGFP backbone + amacGFP features (238 aa)

Strategy: Cross-species chimeric design

Features: sfGFP backbone (proven folding) + amacGFP chromophore environment (brightness) + TGP surface (K79E, Q204E, anti-aggregation) + usGFP core (Q69L, N164Y, stability)

Logic: Most innovative design -- cross-species chimera combining best features from multiple families. sfGFP scaffold for folding, amacGFP-derived environment for brightness, TGP surface for thermostability, usGFP core for packing.

Summary

Seq_ID	Design	Scaffold	Len	Strategy
default_1	sfGFP_TGP	sfGFP	238	Surface engineering
default_2	amacGFP_v1	amacGFP	238	Minimal FoldX (3)
default_3	amacGFP_v2	amacGFP	238	Moderate FoldX (7)
default_4	ppluGFP_surface	ppluGFP	222	Surface engineering
default_5	cgreGFP_v2	cgreGFP	235	Brightness combo
default_6	Chimera	sfGFP+amac	238	Cross-species chimera

Key References

1. Pedelacq et al. 2006. Nat Biotechnol 24:79-88 (sfGFP)
2. Close et al. 2015. Proteins 83:1225-1237 (TGP surface engineering)
3. Scott et al. 2018. Sci Rep 8:159 (usGFP core packing)
4. Sarkisyan et al. 2016. Nature 533:397-401 (avGFP fitness landscape)
5. Sokalingam et al. 2012. PLoS ONE 7:e40410 (Lys->Arg mutagenesis)